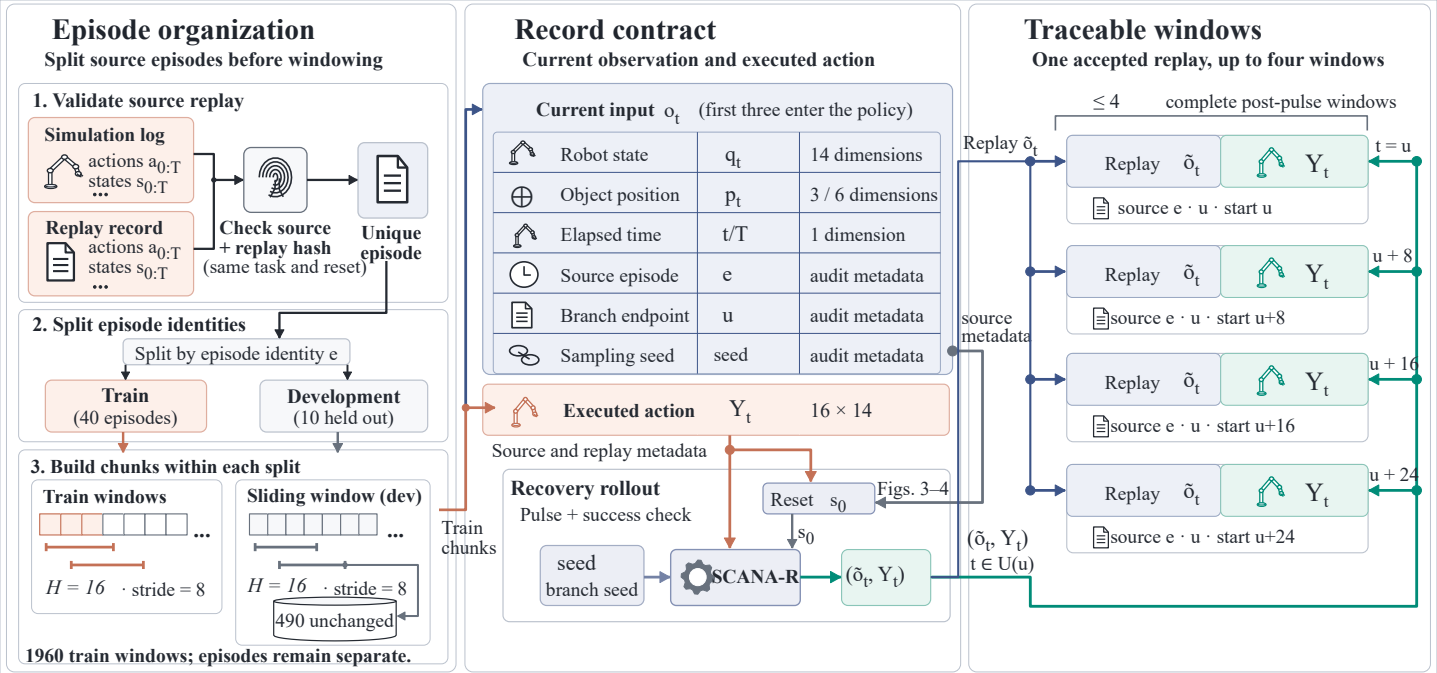

Episode organization
Split source episodes before windowing
1. Validate source replay
Simulation log
actions $a_{0:T}$
states $s_{0:T}$
Replay record
actions $a_{0:T}$
states $s_{0:T}$
Check source + replay hash
(same task and reset)
Unique episode
2. Split episode identities
Split by episode identity e
Train
(40 episodes)
Development
(10 held out)
3. Build chunks within each split
Train windows
H = 16 · stride = 8
Sliding window (dev)
H = 16 · stride = 8
490 unchanged
1960 train windows; episodes remain separate.
Record contract
Current observation and executed action
Current input $o_t$ (first three enter the policy)
Robot state $q_t$ 14 dimensions
Object position $p_t$ 3 / 6 dimensions
Elapsed time t/T 1 dimension
Source episode e audit metadata
Branch endpoint u audit metadata
Sampling seed seed audit metadata
Executed action $Y_t$ 16 × 14
Source and replay metadata
Recovery rollout
Pulse + success check
Reset $s_0$
$s_0$
Figs. 3–4
Train chunks
seed
branch seed
SCANA-R
$(\tilde{o}_t, Y_t)$
Traceable windows
One accepted replay, up to four windows
≤ 4 complete post-pulse windows
Replay $\tilde{o}_t$
source metadata
Replay $\tilde{o}_t$ $Y_t$
source e · u · start u
t = u
source e · u · start u+8
u + 8
source e · u · start u+16
u + 16
source e · u · start u+24
u + 24
$(\tilde{o}_t, Y_t)$
t ∈ U(u)